C-AR.4.5 · Re-check · 3 questions

MA.7.AR.4.5 · Real-world proportional problems · Unit 6 Multi-Step Problems With Proportional Relationships · Given Wednesday, after the reteach, when this benchmark was retaught

Form title C-AR.4.5 Re-check · Real-world proportional problems
Description 3 questions, 1 point each. Work each one out on paper, then choose or type your answer. Calculator allowed.
Locked mode OFF (calculator allowed; Locked mode blocks the Chromebook calculator)

Short-answer validation (already on the template's short-answer question; copy that question): Regular expression → Matches → `^-?(\d+(\.\d+)?|\.\d+|\d+/\d+|\d+ \d+/\d+)$`
Error text: Type only a number, like 12 or -3.5 or 2/7 or 4 1/3. No units, $, %, π or commas.

#	Question image	File · type	Options to add	Answer key	Pts
1	A car travels 23.1 miles on each gallon of gas, and its tank holds 17.5 gallons. The driver starts with an empty tank and fills it completely every time it is filled. What is the least number of times the tank must be filled to drive 1,407 miles? Type a number only. No units, $ or %.	**Q1.png** Short answer	none (typed answer)	**4 · 4.0 · 4.00** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1
2	Two phone plans each charge a cost that is proportional to the number of months. Plan X: the table shows the cost for some numbers of months. Months / Cost: 2 / $55.50; 6 / $166.50; 10 / $277.50 Plan Y: the cost c, in dollars, for m months is given by $c = 28.5m$. Which plan costs less for 12 months, and by how much? A. Plan X, by $9.00 B. Plan X, by $64.50 C. Plan X, by $0.75 D. Plan Y, by $324.00	**Q2.png** Multiple choice Same problem, now multiple choice (paper: a written answer)	A B C D	**A**	1
3	Two machines each fill bottles at a constant rate. Machine P fills 36 bottles in 1.5 minutes. Machine Q's number of bottles, b, after t minutes is given by $b = 22t$. How many more bottles does the faster machine fill in 20 minutes? Type only the number. Type a number only. No units, $ or %.	**Q3.png** Short answer	none (typed answer)	**40 · 40.0 · 40.00** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1

Answers with worked solutions are in the Answers folder next to the question images (same file names).